



Mid term

End Term Examination 2025-26, B. Tech (Semester-II)

Paper Code: 22CST152

Maximum Marks: 30

Maximum Time: 90 Minutes

Note:

Note: Information provided in the questions is assumed to be sufficient, and **no doubts shall be entertained during the examination**. Suitable and/or proper assumption(s) may be considered, if applicable.

Answer the following questions. Answers must be precise to the points.

Q1.(a) Implement the following boolean functions $f_{a,b,c} = \sum(0,6)$ and $f_{a,b,c,d,e} = (a+b)(c+d)e$ using NAND (AND-Invert only) and NOR (OR-Invert only) gates, respectively. [3+3]

(b) Solve/evaluate/prove the following.

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 (i) $(3245)_{10} = (?)_3$, (ii) If $(xy)_{10} = 50$ and $(yx)_2 = 3$, then x, y ?, (iii) $(69-2)_p$ using $(p-1)$'s complement for suitably chosen minimized value of p , (iv) $x+x=x$ and $x.x=x$ (using basic postulates) [2x4=8]

Q2.(a) Find out the boolean function for BCD to Excess-3 code conversion.
(hint: excess-3(0110)=1001)

(b) Find out the essential prime implicant(s) for the following function using K-map method
 $f = \sum(1, 4, 6, 7, 8, 9, 10, 11, 15)$

Q3. Given the following function, find out the corresponding truth table. Also, deduce the function in POS form, starting from considering minterms only.

$$f_{a,b,c,d} = a'd + ac + cb' + bd + ab'd'$$



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मालवीय राष्ट्रीय प्रौद्योगिकी संस्थान जयपुर (राष्ट्रीय महत्व का संस्थान)
Malaviya National Institute of Technology Jaipur
(An Institute of National Importance)
Mid Term Examination 2024-25, B. Tech (Semester-II)

Paper Name: Logic System Design
Paper Code: 22CST152

Maximum Marks: 30

Maximum Time: 90 minutes

Note:

Information provided in the questions are assumed to be sufficient and **no doubts shall be entertained during the examination**. Suitable and/or proper assumption(s) may be taken into consideration, if applicable.

Answer the following questions. All questions are compulsory.

Q1. (a) Distinguish/Identify essential prime implicants for the following function and also justify your answer.

$$f_{(a,b,c,d)} = \sum(0,2,3,4,5,6,7,11,13) \quad [3+2]$$

(b) Compute the following operations in p's complement representation.

$$(i) (33-71)_{r=10, p=r} \quad (ii) (AB-B8)_{r=16, p=r} \quad (iii) (1101-1110)_{r=2, p=r} \quad (iv) (27-41)_{r=8, p=r} \quad (v) (23-11)_{r=4, p=r} \quad [1 \times 5 = 5]$$

[where in the expression (A-B)_{r,p}, A, B are operands and x, y are base of the number system and asking complement, respectively.]

Q2. (a) State the truth table for odd parity of messages of length 4 bits including the parity bit. Design the corresponding circuit using one of the universal gates for the parity function computed. [2+3]

(b) List all decimal numbers/digits in 84-2-1 and 2421 coding schemes. Compute the following operations.

$$(i) (27)_{10} = (?)_{bcd} = (?)_{ex-3} \quad (ii) (00100111)_2 = (?)_{ex-3} \quad [3+2]$$

Q3. (a) Minimize the following boolean function.

$$f = (w+x+y'+z)(w+x+y'+z')(w+x'+y+z)(w+x'+y+z')(w'+x'+y'+z)(w'+x'+y'+z'), \text{ and also derive } f'. \quad [3+2]$$

(b) State the BCD addition algorithm and illustrate with a suitable example. [3+2=5]

-X-X-X-